arXiv:2502.01540v1 [cs.CL] 3 Feb 2025

What is a Number, That a Large Language Model May Know It?

Raja Marjieh [*1] **Veniamin Veselovsky** [*2] **Thomas L. Griffiths**[† 1 2] **Ilia Sucholutsky**[† 3]

Abstract

Numbers are a basic part of how humans represent and describe the world around them. As a consequence, learning effective representations of numbers is critical for the success of large language models as they become more integrated into everyday decisions. However, these models face a challenge: depending on context, the same sequence of digit tokens, e.g., 911, can be treated as a number or as a string. What kind of representations arise from this duality, and what are its downstream implications? Using a similarity-based prompting technique from cognitive science, we show that LLMs learn representational spaces that blend string-like and numerical representations. In particular, we show that elicited similarity judgments from these models over integer pairs can be captured by a combination of Levenshtein edit distance and numerical Log-Linear distance, suggesting an entangled representation. In a series of experiments we show how this entanglement is reflected in the latent embeddings, how it can be reduced but not entirely eliminated by context, and how it can propagate into a realistic decision scenario. These results shed light on a representational tension in transformer models that must learn what a number is from text input.

1. Introduction

> "What is a number, that a man may know it, and a man, that he may know a number?"
>
> *Warren McCulloch* (1961)

Numbers play a pivotal role in many aspects of human cognition (Dehaene, 2011). Starting from a narrow sense of magnitude in infancy, humans gradually acquire nuanced representations of number that capture abstract properties such as "even" and "odd" (Miller & Gelman, 1983; Piantadosi et al., 2014). Decades of research have been devoted to understanding how humans process and represent numbers (Miller & Gelman, 1983; Nieder & Dehaene, 2009; Cheyette & Piantadosi, 2020; Tenenbaum, 1999; Piantadosi, 2016). In fact, this endeavor dates back to some of the earliest applications of neural networks to artificial intelligence (McCulloch & Pitts, 1943; McCulloch, 1961) when researchers sought to define computing machines (a simulated "man") that can intelligently process numbers.

The question of how computing machines represent number assumes key importance as modern neural networks, in particular large language models (LLMs), are becoming integrated into everyday decisions (Zhu et al., 2025; Hanna et al., 2023; Stolfo et al., 2023; McCoy et al., 2024a;b; Tian et al., 2023). Since these models learn representations by predicting tokens in text, LLMs face a challenge: depending on context, the same sequence of digit tokens, e.g., 101 or 911, can be treated as a number or as a string. This duality introduces an issue akin to polysemy and homonymy (Vicente & Falkum, 2017), but it extends across different symbolic systems. What kind of representations arise from this duality, and what are its downstream implications? This is not straightforward to answer, in part because LLMs lack the experience that shapes humans' number representations, and in part because model internals are not always available and diagnostic tests can be challenging to design.

Here we address this challenge by leveraging tools from cognitive science for characterizing representations (Shepard, 1980; 1987; Tversky & Hutchinson, 1986; Tenenbaum & Griffiths, 2001; Marjieh et al., 2024a). Specifically, we elicit similarity judgments across number pairs from six modern LLMs using an appropriate prompt. We then use those judgments to construct detailed maps (similarity matrices) that capture how the models organize numbers. Crucially, this technique can applied to any model without the need to access its internals. Moreover, when the model internals are available, this approach can be combined with probing techniques to evaluate how prompt behavior is reflected in the structure of the latent embeddings.

We show that despite variation in model training data and size, all models exhibit highly regular patterns that can be adequately decomposed in terms of a psychologically-motivated numerical distance based on a Log-Linear rep-

[†*]Equal contribution [1]Department of Psychology, Princeton University [2]Department of Computer Science, Princeton University [3]Center for Data Science, New York University. Correspondence to: Raja Marjieh <raja.marjieh@princeton.edu>.

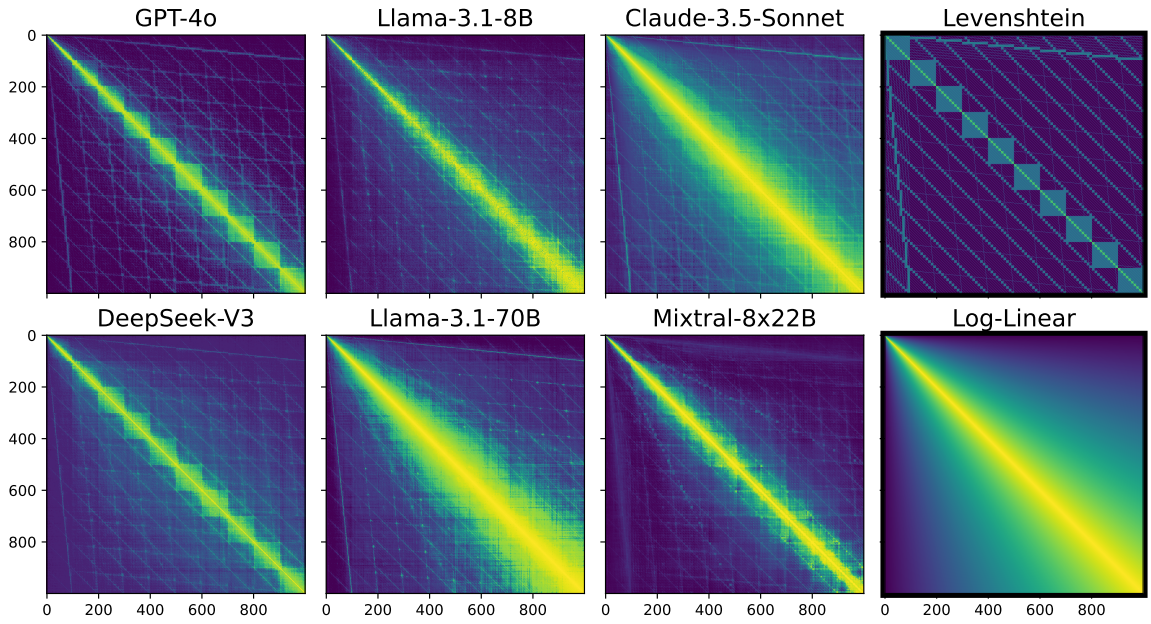

Figure 1. LLM number similarity matrices (symmetrized) over all integer pairs in the range $0 - 999$, along with two theoretical similarity matrices derived from a Levenshtein string edit distance and a psychological Log-Linear numerical distance (highlighted in black).

resentation of numbers (Piantadosi, 2016) and a string edit distance, suggesting an entanglement. We show how this entanglement can be reduced but not eliminated by introducing a context that specifies the 'type' of the integer (`int()` vs. `str()`), and how it can be probed internally in a model's latent space. Inspired by these results we then construct a decision scenario that reveals how string-bias can propagate into a realistic setting leading to incorrect answers. Viewed together, these findings shed light on an intrinsic tension between number and string representations that modern large language models must learn to navigate.

2. Related Work

2.1. Numbers and Language Models

The processing and representation of numbers in large language models is a topic of active research. Number representations have recently been studied within pretrained language models using techniques from mechanistic interpretability. For example, Zhu et al. (2025) devised a dataset of addition problems to show that LLMs encode the value of numbers linearly in that context. Likewise, circuit analysis has been adopted to characterize how smaller language models compute "greater than" (Hanna et al., 2023) and basic arithmetic operations (Stolfo et al., 2023). Other work studied how simple models solve modular arithmetic tasks (Nanda et al., 2023a). More behavioral approaches have also been explored. Tian et al. (2023) looked at how model generations of numbers can be used to test for answer calibration. Likewise, McCoy et al. (2024a; 2024b) showed how a Bayesian framework can be used to characterize how LLM performance on various tasks, including numeric ones, depends on the probability of the input, task, and output.

2.2. Probing and Representation Analysis in LLMs

Probing is a well-established technique for studying the internal representations of machine learning models (Alain & Bengio, 2018; Belinkov, 2021). Dating back to BERT (Devlin et al., 2019), probing has been used to explore various aspects of representation such as how language models encode sentence structure (Tenney et al., 2019; Manning et al., 2020; Rogers et al., 2021). More recently, probing has become a common technique in studying larger language models. In the work by Zhu et al. (2025) mentioned earlier, the authors designed linear probes to extract number value from latent embeddings. Probes have also been used to discover how language models process concepts across layers (Gurnee et al., 2023), and for extracting uncertainty in language model generations (Kadavath et al., 2022). Similarly, probes, and latent analysis more broadly, have been used in the world models literature, e.g., to show that language models trained on Othello games develop a world model of the game (Nanda et al., 2023b), and that they learn to represent space and time (Gurnee & Tegmark, 2024; Gurnee et al., 2023; Nylund et al., 2023). More recent approaches to studying representations in language models have also been proposed like activation patching (Meng et al., 2022), sparse autoencoders (Cunningham et al., 2023;

Gao et al., 2024), and distributed alignment search (Geiger et al., 2024).

2.3. Behavioral Analysis of Language Models

By combining the prompt comprehension capabilities of large language models with the wide range of paradigms available in the behavioral sciences, a growing line of work proposes new tools for characterizing behavior in LLMs. For example, Marjieh et al. (2024b) used a suite of perceptual judgment tasks from psychophysics to characterize sensory knowledge in LLMs. Binz & Schulz (2023) leveraged paradigms from cognitive psychology to study how GPT-3 solves various tasks. Bai et al. (2024) used tools from social psychology to diagnose implicit racial and gender bias in LLMs. Webb et al. (2023) subjected LLMs to analogical tasks to study their abstract reasoning capacities. Our work takes a similar approach to this line of work, and combines it with modern probing techniques to study LLM behavior and representation in the domain of number.

3. Probing Number Representations in LLMs with Similarity Judgments

Our approach builds on the paradigm of similarity judgments from cognitive science (Shepard, 1962; 1980; Tversky & Hutchinson, 1986; Tenenbaum & Griffiths, 2001; Marjieh et al., 2024a), which is also closely related to representational similarity analysis (RSA) from neuroscience (Kriegeskorte et al., 2008). Given a domain of interest $\mathcal{D}$, a set of representative items from that domain $\{x_1, x_2, \ldots, x_N\} \subset \mathcal{D}$ (or 'stimuli'), and an agent $\mathcal{A}$ whose representation $\mathcal{M}(\mathcal{D})$ one would like to characterize, the paradigm proceeds by eliciting pairwise similarity judgments from $\mathcal{A}$ across all pairs of items ('How similar is the item x_i to the item x_j?'). These judgments are then aggregated into a similarity matrix s_{ij} which defines a notion of proximity on $\mathcal{D}$ that can be used to characterize $\mathcal{M}(\mathcal{D})$. Since the notion of similarity is by construction neutral, it leaves it to the agent to impose its own structure on s_{ij}.

In our case, the domain of interest is the set of non-negative integers $\{0, 1, 2, \ldots\}$, and the agent is the LLM in question. We consider a representative sample of state-of-the-art models, namely, GPT-4o (Hurst et al., 2024), two variants of Llama-3.1 (8b and 70b) (Dubey et al., 2024), DeepSeek-V3 (Liu et al., 2024), Claude-3.5-Sonnet (Anthropic, 2024), and Mixtral-8x22B (Jiang et al., 2024). We then apply the similarity technique in a series of experiments that span different contexts as well as behavioral (prompt-level) and internal (embedding-level) analyses and examine how the observed patterns decompose in terms of theoretical string and numerical metrics. We detail those experiments in what follows.

4. Methodology

4.1. Tasks

Our experiments fall into three categories, all of which are associated with a certain behavioral prompt for eliciting LLM judgments over numerical quantities. The prompts are provided in Appendix A. In the first category, we elicited similarity judgments over all pairs of numbers in the range $0 - 999$ ('How similar are the two numbers on a scale of 0 (completely dissimilar) to 1 (completely similar)?'; see Appendix A). We then constructed symmetric similarity matrices s_{ij} by averaging over both presentation orders (i.e., $s(x_i, x_j)$ and $s(x_j, x_i)$), and then evaluated the structure that emerged (see Evaluation Metrics below) and the way it is affected by context. We focused on the range $0 - 999$ because all integers within it are represented as unique tokens in the models considered, which controls for any tokenization-specific differences. In the second category, we extended the similarity analysis for one of the models for which we had internal access (Llama-3.1-8b) and probed the extent to which its internal token embeddings reflected the external behavior through a linear transformation (see Probing Language Models). Finally, in the third category we constructed a naturalistic decision scenario in which the model had to select from two available numerical quantities q_1, q_2 (a compound concentration in a test tube) the one that is most similar to a desired quantity q_0 (further details in Experiments and Appendix A). We constructed such triplets (q_0, q_1, q_2) in a way that the answer would diverge depending on whether the quantities are treated as numbers or strings (see Constructing Close Triplets below) which then allowed us to probe the string-bias of models.

4.2. Models

We evaluated four open-source and two closed-source models. The open source models consist of Llama-3.1-8b-Instruct, Llama-3.1-70b-Instruct, DeepSeek-V3, Mixtral-8x22B-Instruct-V0.1, and the closed source include GPT-4o and Claude-3.5-Sonnet. Henceforth, we will refer to the Llama and Mixtral models as Llama-3.1-8b, Llama-3.1-70b, and Mixtral-8x22B to improve readability. Since generating 1000×1000 similarities requires sampling 1,000,000 pairwise comparisons which costs around $160 for Claude-3.5-Sonnet, we limit the model evaluations across all contexts to one run at a temperature of zero to get the most likely answer across the different settings. For comparison, we include results from an additional run in the basic similarity context at a temperature of 0.7 in Appendix F.

4.3. Evaluation Metrics

To characterize the extent to which the derived similarity matrices s_{ij} were string-like or number-like, we regressed their